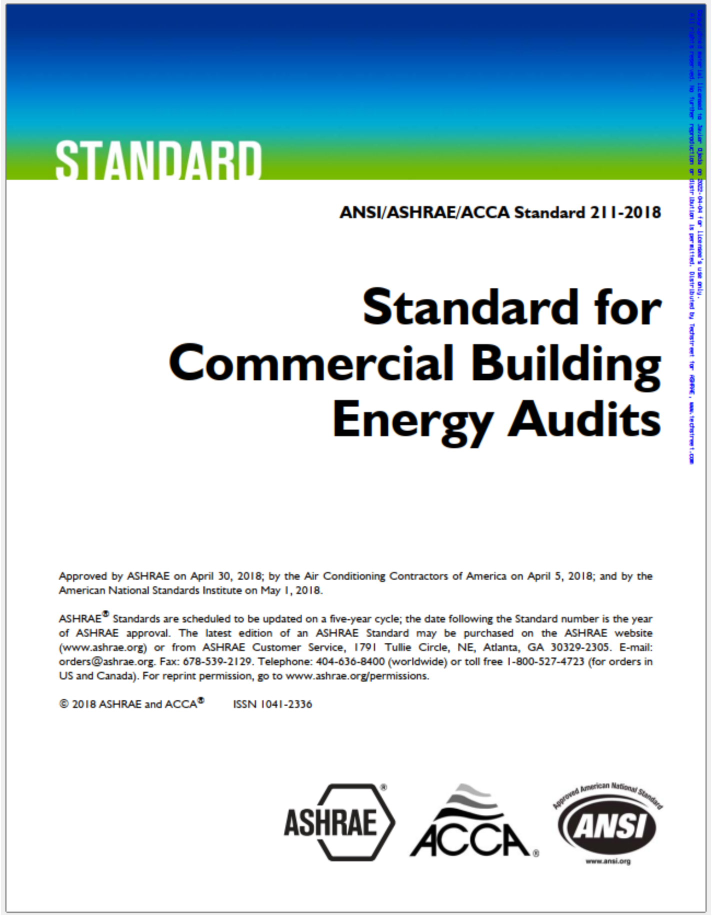
Agenda

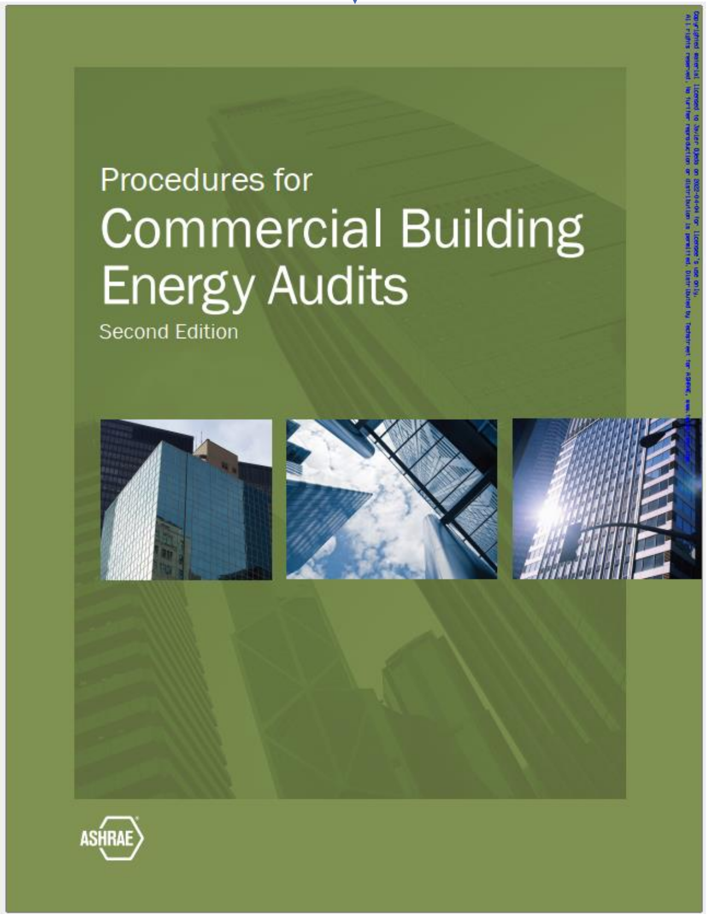
1. Introduction
   - What's ASHRAE 211-2018
   - What's RETSCreen
2. Planning the Audit
   - Audit Scope and Objectives
   - Info that you will need
   - Preliminary Energy Use Analysis
   - Tools
   - Team
3. Conducting the Audit
   - Site Visits
   - Data Analysis – RETScreen + Excel + ASHRAE Forms
   - Report Structure
4. After the audit
   - RETScreen Energy Modeling
   - Model Calibration
   - EEM's and their simulation in RETScreen
5. Final Report

Introduction

This is the PCBEA!

- The ASHRAE 211-2018 Standard
  - A complement to the **PCBEA**
  - Brings order into chaos of Energy Audits
  - Establishes energy audit levels 1, 2, 3
  - Establishes Preliminary Energy Use Analysis Methods
  - Establishes a minimum performance standard for energy audits





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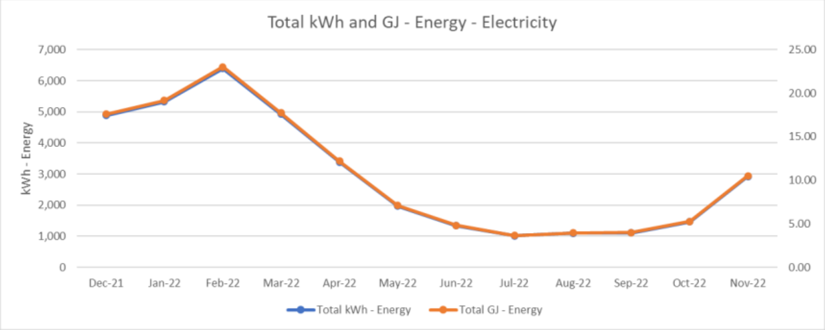
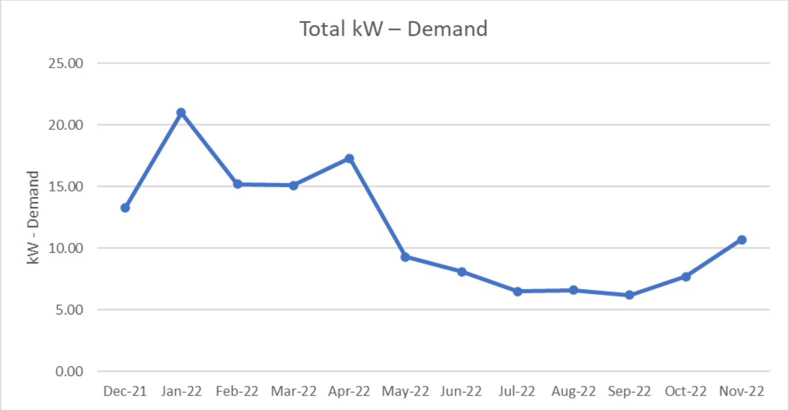
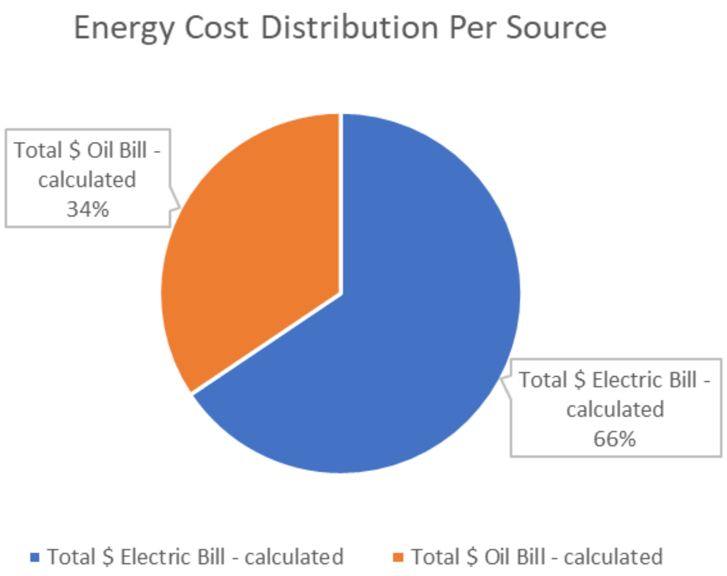
Planning the Audit

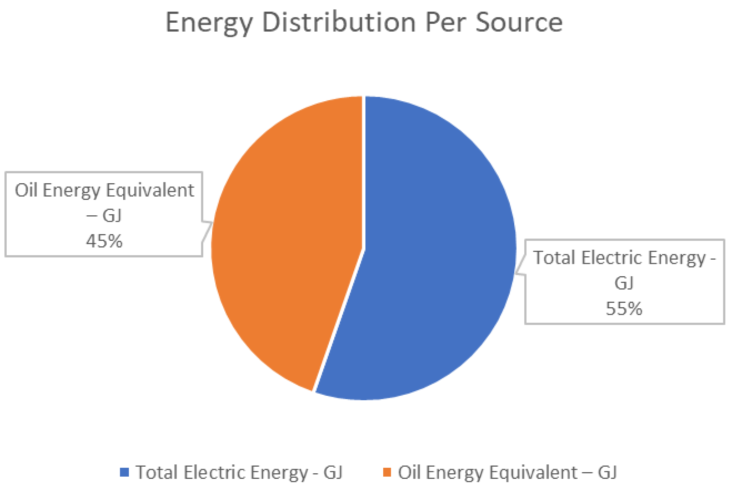
Preliminary Energy Use Analysis:

- Baseline overview of building's energy performance
- Early Stage Benchmarking
- Data Sources: utility bills, schedules (min. 1 calendar year)
- Key Metrics:
  - EUI
  - ECI
- Informs depth & focus of main audit









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RETScreen Expert

ENERGY STAR PortfolioManager®

ASHRAE Free Templates

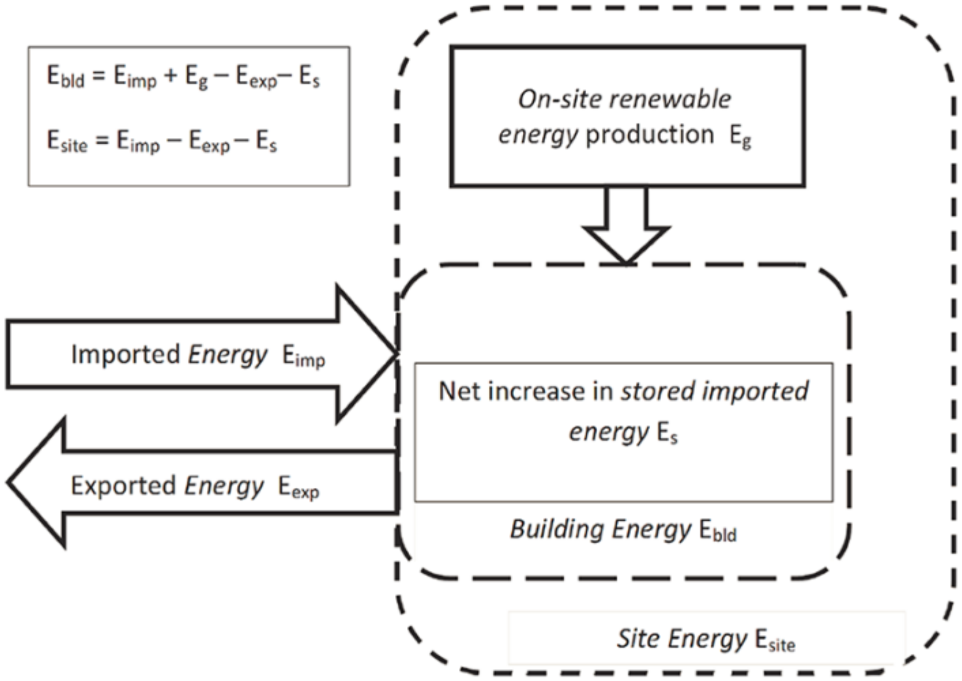


Figure 1 Relationship of site energy (E_{site}) and building energy (E_{bld}). From ASHRAE 211-2018, page 5

EUI: Energy Use Intensity

$$\frac{total\ energy\ consumed\ by\ the\ building\ in\ one\ year\ (measured\ in\ kBtu\ or\ GJ)}{total\ gross\ floor\ area\ of\ the\ building\ (measured\ in\ square\ feet\ or\ square\ meters)} = \left(\frac{Kbtu}{SQft}\right)\ or\ \left(\frac{GJ}{SQM}\right)$$

ECI: Energy Cost Index

$$\frac{total\ cost\ energy\ consumed\ by\ the\ building\ in\ one\ year\ (measured\ in\ \$)}{total\ gross\ floor\ area\ of\ the\ building\ (measured\ in\ square\ feet\ or\ square\ meters)} = \left(\frac{\$}{SQft}\right)\ or\ \left(\frac{\$}{SQM}\right)$$

Conducting the Audit – Report Structure

1. Executive Summary

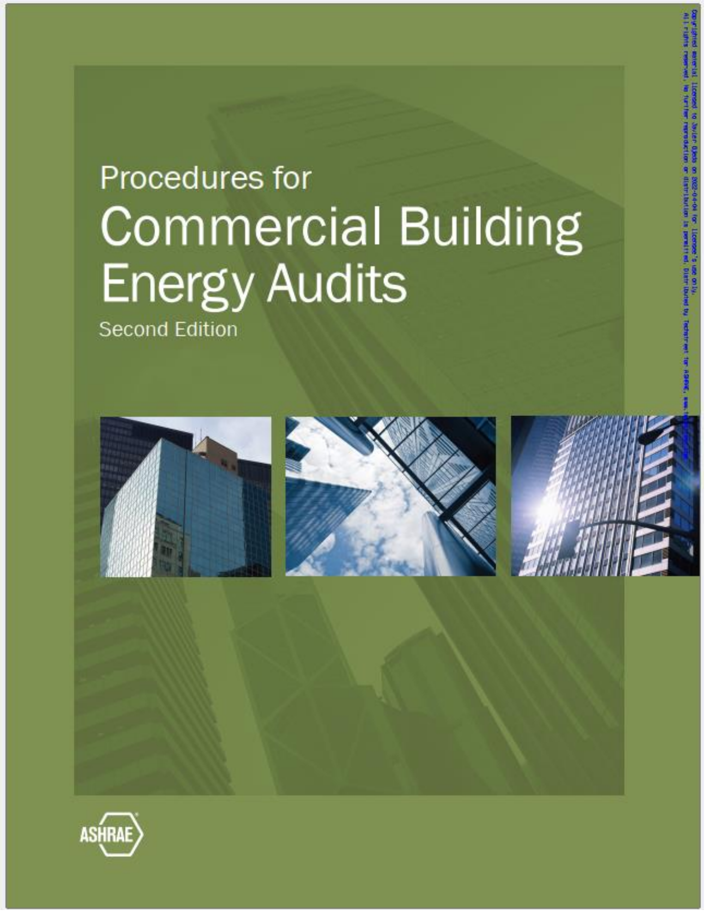
- Brief background and summary of scope
- EEM (Energy Efficiency Measures) summary table
- Summary of benchmarking results

2. Background Information

- Contact information for stakeholders
- Audit scope, methodology
- Description of the site and the building
- Historical energy consumption and costs
- Benchmarking results
- End-use analysis results
- Client motivations or mandates satisfied by this effort

3. Description of the Existing Building Systems

- Occupancy
- Building envelope
- Lighting systems
- Mechanical systems
- Security, oversight, or operational requirements impacting this effort
- Photographs



4. Energy Efficiency Measures

- No-cost/low-cost measures
- Capital investment measures
- Demand response measures
- Distributed generation/renewable energy measures
- O&M measures
- Impact of current financial market, utility incentives, federal tax or grant support, etc.

5. Supporting Information

- Analysis
- Measured data or monitoring results
- Manufacturers' information or "cut sheets"
- Plans and sketches
- Additional information or specifications

Let's go with Live Example!